The Moving Earth: Understanding Landslide Events

MIS 381N: Information Management - Fall 2025

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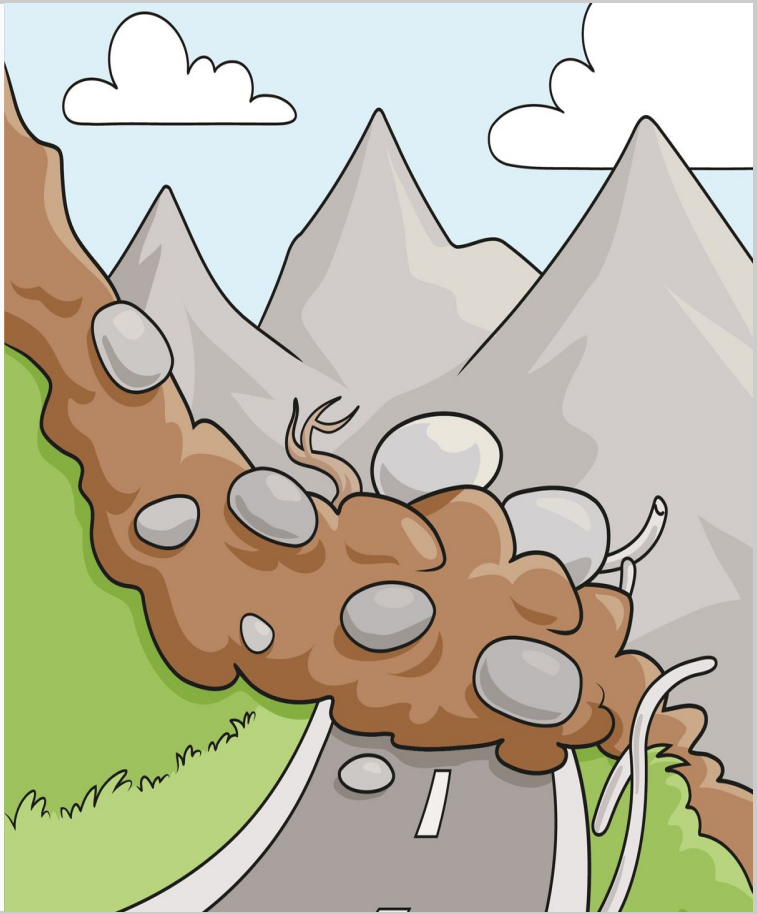
Agenda

- **Dataset Overview**
  NASA's Global Landslide Catalog
- **Architecture & Schema**
  Medallion Model & Star Schema Design
- **Business Use Cases**
  Gold Layer Reporting & Steamlit
- **AI & Advanced Analytics**
  Cortex Functions & Predictive Insights



Dataset Overview

The Global Landslide Catalog (GLC)
A comprehensive dataset maintained by NASA Goddard Space Flight Center, cataloging rainfall-triggered landslide events globally.

- Source: NASA / Kaggle
- Volume: 11,000+ recorded events
- Key Attributes: 31 attributes including Event Title, Event Description, Date, Location (Lat/Long), Trigger Type, Fatalities, Landslide Size, etc.
- Goal: Identify high-risk zones and trigger patterns.

Medallion Architecture

We implemented a robust 3-layer architecture within Snowflake to ensure data quality and performance.

- **Bronze (Raw):** Direct ingestion of the Kaggle CSV dataset into the LANDSLIDE_RAW table. No transformations.
- **Silver (Refined):** Cleaned, normalized, and structured into a Star Schema. Deduplication and type casting applied.
- **Gold (Curated):** Aggregated business-level tables optimized for reporting (e.g., QUATERLY_LANSDLIDE_FATALITIES).

Silver Layer: Star Schema Design

DIM_SOURCE
- PK SOURCE_ID
- SOURCE_NAME

DIM_COUNTRY
- PK COUNTRY_ID
- COUNTRY_NAME
- COUNTRY_CODE

DIM_ADMIN_DIVISION
- PK ADMIN_DIVISION_ID
- ADMIN_DIVISION_NAME

DIM_DISTANCE
- PK DISTANCE_ID
- LOCATION_ACCURACY

DIM_LANDSLIDE_CATEGORY
- PK CATEGORY_ID
- CATEGORY

DIM_LANDSLIDE_TRIGGER
- PK TRIGGER_ID
- LANDSLIDE_TRIGGER

FACT_LANDSLIDE

PK LANDSLIDE_EVENT_KEY

-- Date Keys (FK) --
- EVENT_DATE_ID
- SUBMITTED_DATE_ID
- CREATED_DATE_ID
- LAST_EDITED_DATE_ID

-- Dimension Keys (FK) --
- SOURCE_ID
- COUNTRY_ID
- ADMIN_DIVISION_ID
- DISTANCE_ID
- CATEGORY_ID
- TRIGGER_ID
- GAZETTEER_ID
- SETTING_ID
- SIZE_ID
- STORM_ID

-- Measures & Other --
- FATALITY_COUNT NUM
- INJURY_COUNT NUM
- LONGITUDE NUM
- LATITUDE NUM
- ADMIN_DIVISION_POPULATION NUM
- GAZETTEER_DISTANCE NUM
- EVENT_ID VARCHAR
- EVENT_TITLE VARCHAR
- EVENT_DESCRIPTION VARCHAR
- LOCATION_DESCRIPTION VARCHAR
- NOTES VARCHAR
- PHOTO_LINK VARCHAR
- SOURCE_LINK VARCHAR

DIM_DATE (Role Playing)
- PK DATE_ID
- DATE, YEAR, MONTH, DAY, DAY_OF_WEEK, QUARTER
- *Used for Event, Created, Submitted, Edited dates

DIM_GAZETTEER
- PK GAZETTEER_ID
- GAZETTEER_CLOSEST_POINT

DIM_LANDSLIDE_SETTING
- PK SETTING_ID
- SETTING

DIM_LANDSLIDE_SIZE
- PK SIZE_ID
- LANDSLIDE_SIZE

DIM_STORM
- PK STORM_ID
- STORM_NAME

Streamlit Dashboard

Streamlit Apps Final_Project_Landslides Active Share Run

Packages

```python
import streamlit as st
from snowflake.snowpark.context import get_active_session
import pandas as pd
import altair as alt

# Configuration
st.set_page_config(page_title="Global Landslide Analytics",

# Get the active Snowflake session
session = get_active_session()

st.title("Global Landslide Dashboard")
st.markdown("Analysis of landslide events sourced from **DE

# Load data from Gold layer
@st.cache_data
def load_geo_data():
    """
    Reads from DB_TEAM_CLAWJAWCOIL.GOLD.COUNTRY_FATALITIES
    """
    query = "SELECT * FROM DB_TEAM_CLAWJAWCOIL.GOLD.COUNTRY
    return session.sql(query).to_pandas()

@st.cache_data
def load_trigger_data():
    """
    Reads from DB_TEAM_CLAWJAWCOIL.GOLD.TRIGGER_SIZE_FATAL
    """
    query = "SELECT * FROM DB_TEAM_CLAWJAWCOIL.GOLD.TRIGGEF
    return session.sql(query).to_pandas()

@st.cache_data
def load_monthly_data():
    """
    Reads from DB_TEAM_CLAWJAWCOIL.GOLD.MONTHLY_LANDSLIDES_
    """
```

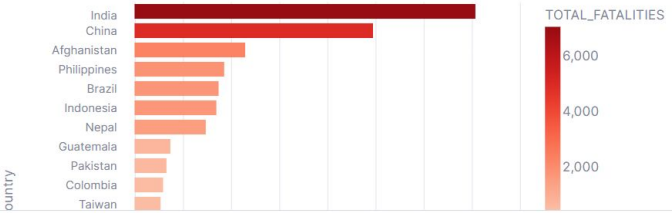
Global Landslide Dashboard

Analysis of landslide events sourced from **DB_TEAM_CLAWJAWCOIL** (Gold Layer).

Total Fatalities	Total Recorded Events	Most Impacted Country
28,898	11,038	India

Geographic Analysis

Fatalities by Country



Fetching app info... Feedback

From the Gold Layer Tables

COUNTRY	TOTAL_FATALITIES
India	7069

Left: COUNTRY_FATALITIES
Below: TRIGGER_SIZE_FATALITIES

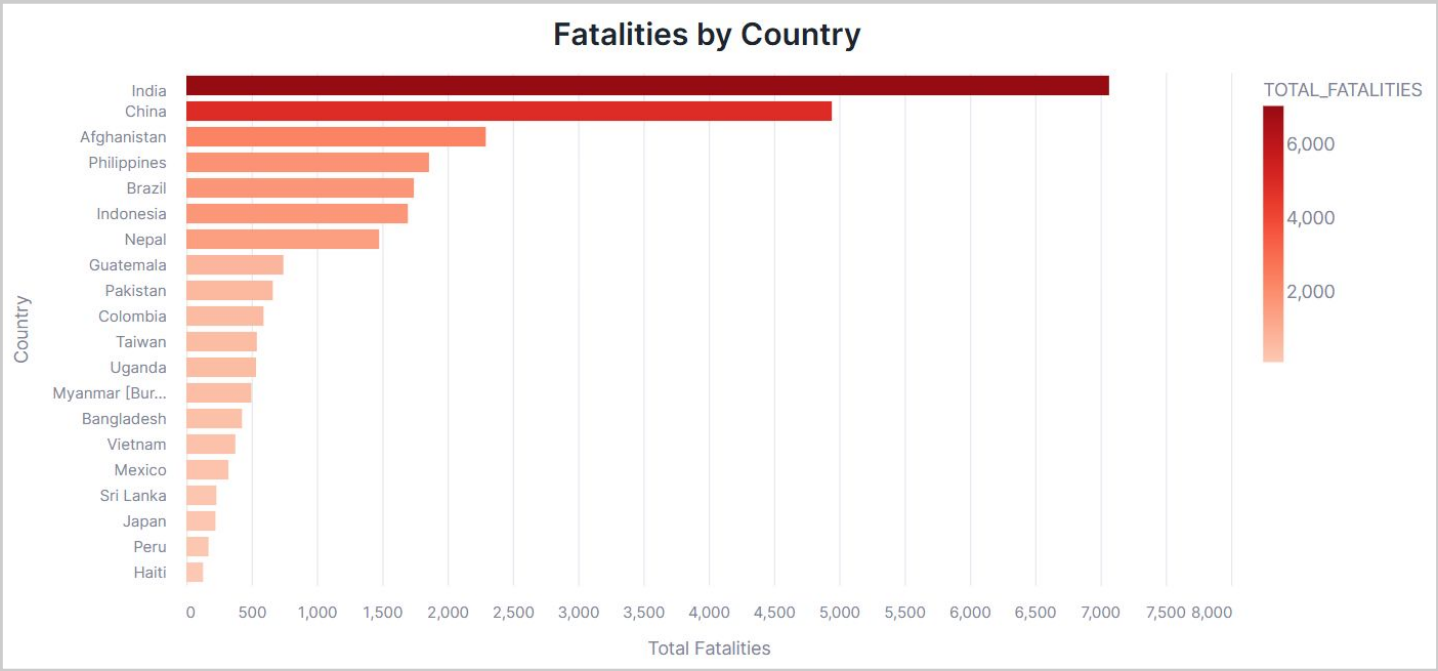
LANDSLIDE_TRIGGER	LANDSLIDE_SIZE	NUM_LANDSLIDES	TOTAL_FATALITIES
tropical_cyclone	Medium	458	1050

MONTH	MONTH_NAME	NUM_LANDSLIDES	TOTAL_FATALITIES
1	Jan	968	1994

Above: MONTHLY_LANDSLIDES_FATALITIES
Right: QUARTERLY_LANDSLIDE_FATALITIES

QUARTER	NUM_LANDSLIDES	TOTAL_FATALITIES
Q1	2760	4486

1. Which countries have the most fatalities?



<u>Top 5 Total Fatalities:</u>

1. India (7,069)
2. China (4,945)
3. Afghanistan (2,294)
4. Philippines (1,859)
5. Brazil (1,743)

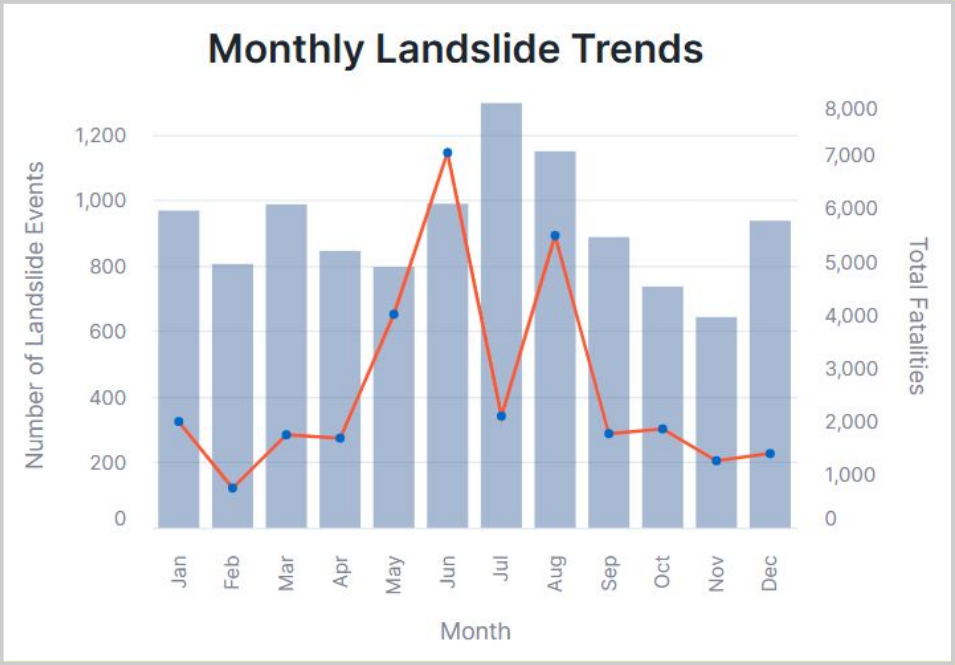
<u>Note:</u> Cleaned from 2,183 rows of countries unknown to 796 rows of countries unknown. (11,000+ total rows)

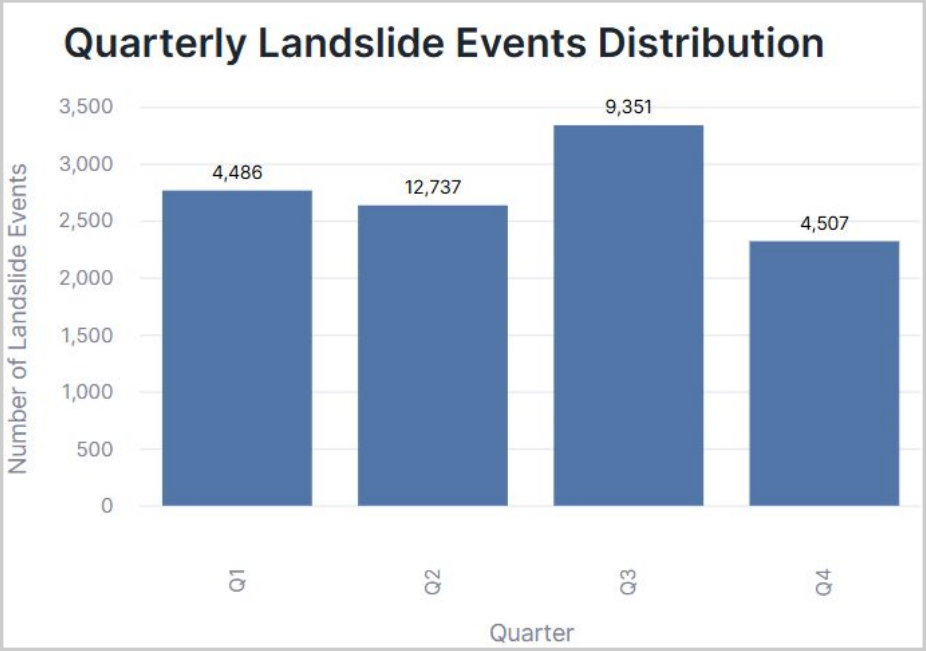
2. What are the most common triggers and their severity?

EVENT_TYPE	TOTAL_FATALITIES
downpour (Large)	13363
downpour (Medium)	3846
continuous_rain (Large)	3101
rain (Medium)	2343
tropical_cyclone (Large)	1819
rain (Large)	1183
tropical_cyclone (Medium)	1050
monsoon (Large)	451

<u>Note:</u> Shown are trigger/size categories with more than 450 total fatalities.

3. What are the seasonal patterns in landslide occurrences?





AI-SQL : Enriching Event Descriptions

- Target table : BRONZE.LANDSLIDE_RAW
- Text Column : EVENT_DESCRIPTION
- AI-SQL functions applied:
  - AI_SENTIMENT(EVENT_DESCRIPTION) → DESCRIPTION_SENTIMENT
  - SNOWFLAKE.CORTEX.SUMMARIZE(EVENT_DESCRIPTION) → DESCRIPTION_SUMMARY
- Output stored as additional columns on the Bronze table

From Raw Descriptions to AI-Generated Signals

	EVENT_ID	EVENT_DESCRIPTION	DESCRIPTION_SENTIMENT	DESCRIPTION_SUMMARY
0	9999903	Heavy rains triggered a mudslide, blocking SR 20.	{ "categories": [{ "name": "overall", "sentiment": "neutral" }]	Heavy rains triggered a mudslide, blocking SR 20.
1	9999904	Melting glacier caused a debris flow.	{ "categories": [{ "name": "overall", "sentiment": "neutral" }]	Melting glacier caused a debris flow.
2	9999905	Typhoon causes multiple small landslides.	{ "categories": [{ "name": "overall", "sentiment": "neutral" }]	Typhoon causes multiple small landslides.
3	9999906	A 6.2M earthquake triggered a rockfall.	{ "categories": [{ "name": "overall", "sentiment": "neutral" }]	A 6.2M earthquake triggered a rockfall.
4	9999907	Continuous monsoon rain led to slide.	{ "categories": [{ "name": "overall", "sentiment": "neutral" }]	Continuous monsoon rain led to slide.
5	684	occurred early in morning, 11 villagers buried in 7 houses	{ "categories": [{ "name": "overall", "sentiment": "negative" }	occurred early in morning, 11 villagers buried in 7 house
6	956	Hours of heavy rain are to blame for an overnight mudslide in Lake Oswego	{ "categories": [{ "name": "overall", "sentiment": "neutral" }]	Hours of heavy rain are to blame for an overnight mudsl
7	973	(CBS/AP) At least 10 people died and as many as 80 were still missing Wed	{ "categories": [{ "name": "overall", "sentiment": "negative" }	At least 10 people died and up to 80 were missing due t
8	1067	One person was killed in Dailekh district, police said.	{ "categories": [{ "name": "overall", "sentiment": "negative" }	One person was killed in Dailekh district, police said.
9	2603	Another landslide in sitio Bakilid in barangay Lahug also left two families ho	{ "categories": [{ "name": "overall", "sentiment": "negative" }	Two families were left homeless by a landslide in sitio B

This enrichment makes the descriptive column easier to filter, search, and aggregate without losing the original narrative!

Cortex Search: Semantic Search on Descriptions

```
SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
  'DB_TEAM_CLAWJAWCOIL.BRONZE.LANDSLIDE_DESC_SEARCH_SVC',
  '{
     "query": "heavy rain monsoon",
     "columns": ["EVENT_ID","EVENT_DATE","COUNTRY_NAME","LANDSLIDE_TRIGGER","EVENT_DESCRIPTION"],
     "limit": 5
   }'
);
```

	SNOWFLAKE.CORTEX.SEARCH_PREVIEW('DB_TEAM_CLAWJAWCOIL.BF
0	{"results":[{"EVENT_DATE":"9/21/08 0:00","@scores":{"text_match":0.0342

```
SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
  'DB_TEAM_CLAWJAWCOIL.BRONZE.LANDSLIDE_DESC_SEARCH_SVC',
  '{
     "query": "killed buried houses destroyed homeless",
     "columns": ["EVENT_ID","EVENT_DATE","COUNTRY_NAME","LANDSLIDE_TRIGGER","EVENT_DESCRIPTION"],
     "limit": 5
   }'
);
```

	SNOWFLAKE.CORTEX.SEARCH_PREVIEW('DB_TEAM_CLAWJAWCOIL.BF
0	{"results":[{"EVENT_DATE":"8/15/14 0:00","@scores":{"text_match":0.0015

```
SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
  'DB_TEAM_CLAWJAWCOIL.BRONZE.LANDSLIDE_DESC_SEARCH_SVC',
  '{
     "query": "earthquake rockfall road collapse",
     "columns": ["EVENT_ID","EVENT_DATE","COUNTRY_NAME","LANDSLIDE_TRIGGER","EVENT_DESCRIPTION"],
     "limit": 5
   }'
);
```

	SNOWFLAKE.CORTEX.SEARCH_PREVIEW('DB_TEAM_CLAWJAWCOIL.BF
0	{"results":[{"EVENT_DATE":"4/11/17 13:34","@scores":{"text_match":0.0621

Different semantic queries pull out different types of landslide events from the same unstructured description column.

Enhancing Search with a Human-Impact Flag (Rule-Based)

- Optional derived feature for Task 12 :
  - HUMAN_IMPACT_FLAG (BOOLEAN, not an AI function)
  - Set to “TRUE” if “EVENT_DESCRIPTION” contains keywords:
    - “kill”, “dead”, “buried”, “injur”, “evacuat”, “homeless”

```sql
SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
  'DB_TEAM_CLAWJAWCOIL.BRONZE.LANDSLIDE_DESC_SEARCH_SVC',
  '{
     "query": "killed buried houses destroyed homeless",
     "columns": ["EVENT_ID","EVENT_DATE","COUNTRY_NAME","LANDSLIDE_TRIGGER","EVENT_DESCRIPTION"],
     "limit": 5
   }'
);
```

Cortex Analyst : Querying Structured Silver layer

- **Source** : Semantic model on the Silver layer
  - Measures :
    - Total Fatalities
    - Event Count
  - Dimensions :
    - Country
    - Event Date
    - Landslide Trigger, Landslide Category

1. Countries with highest landslide fatalities over time
2. Most common triggers and their severity
3. Seasonal patterns in landslide occurrences

Use Case 1 : Countries with Highest Landslide Fatalities

User

Which countries have the highest total landslide fatalities over time? Show a bar chart of total fatalities by country, sorted from highest to lowest

Cortex Analyst

This is our interpretation of your question:

Which countries have the highest total landslide fatalities over the entire available time period? Show total fatalities by country, sorted from highest to lowest.

COUNTRY_NAME	TOTAL_FATALITIES
India	7069
China	4945
Afghanistan	2294
Unknown	2183
Philippines	1859
Brazil	1743
Indonesia	1697
Nepal	1477
Guatemala	743
Pakistan	662
Colombia	591

- **Why it matters:**
  - **Prioritize disaster-preparedness resources**
  - **Support policy makers in high-risk regions**
  - **Identify vulnerable populations**

Use Case 2 & 3 : Triggers and Seasonal Patterns

User

What are the most common landslide triggers and how severe are they? Show events and fatalities by trigger, plus average fatalities per event."

Cortex Analyst

This is our interpretation of your question:

What are the most common landslide triggers and how severe are they? Show the number of events and total fatalities by trigger type, plus average fatalities per event for each trigger type.

LANDSLIDE_TRIGGER	TOTAL_EVENTS	TOTAL_FATALITIES	AVG_FATALITIES_PER_EVENT
downpour	4681	17338	3.703909
rain	2592	3683	1.420910
unknown	1691	858	0.507392
continuous_rain	748	3572	4.775401
tropical_cyclone	562	2895	5.151246
snowfall_snowmelt	135	492	3.644444
monsoon	130	730	5.615385
mining	93	382	4.107527
earthquake	90	456	5.066667
construction	82	142	1.731707
flooding	75	151	2.013333

User

What seasonal patterns exist in landslide occurrences? Show monthly or seasonal event counts and fatalities.

Cortex Analyst

This is our interpretation of your question:

What seasonal patterns exist in landslide occurrences? Show monthly event counts and fatalities aggregated by month across all available years in the data to identify seasonal trends.

MONTH	TOTAL_EVENTS	TOTAL_FATALITIES
1	968	1994
2	805	746
3	987	1746
4	845	1683
5	796	4010
6	989	7044
7	1296	2097
8	1149	5487
9	887	1767
10	736	1857
11	643	1258

- Distinguish frequent vs high-impact triggers
- Guide warning systems and infrastructure planning

- Forecast risky periods (rainy seasons, monsoon)
- Optimize monitoring and emergency response staffing

Automated Incremental Loading Pipeline

Orchestrating the flow from Bronze to Gold using Snowflake Streams & Procedures

- We simulated incoming data using a secondary file Global_Landslide_Data_New_Dummy.csv

1. INGESTION

Bronze Table
Raw History

Snowflake Stream
Source from Raw ingestion
Captures New Rows Only

New Data Arrives

Triggers Pipeline

2. ORCHESTRATION

Stored Procedure
SP_LOAD_FACT_LANDSLIDE

Step A: Silver Load
Consume Stream & Append

```
INSERT INTO FACT... SELECT FROM STREAM
```

Step B: Gold Refresh
Re-calculate Aggregates

```
CREATE OR REPLACE TABLE GOLD...
```

Data Committed

Updates Views

3. ANALYTICS

Silver Fact
Appended Old + New History

Gold Aggregates
Fresh KPI Totals

Task9_code7

```
-- B. Wait 30 seconds, then verify data is in the stream
SELECT COUNT(*) FROM BRONZE.LANDSLIDE_STREAM;
```

COUNT(*)
5

Incremental Loading Comparison

SQL as Task9_code5_1

```sql
-- 2. Check Current Row Counts (Baseline)
SELECT COUNT(*) AS BRONZE_COUNT FROM BRONZE.LANDSLIDE_RAW;
```

	BRONZE_COUNT
0	11,033

SQL as Task9_code5_2

```sql
SELECT COUNT(*) AS SILVER_FACT_COUNT FROM SILVER.FACT_LANDSLIDE;
```

	SILVER_FACT_COUNT
0	11,033

```sql
-- Check Gold (and/or check a specific metric you changed)
SELECT * FROM GOLD.QUARTERLY_LANDSLIDE_FATALITIES
ORDER BY QUARTER;
```

	QUARTER	NUM_LANDSLIDES	TOTAL_FATALITIES
0	Q1	2,760	4,486
1	Q2	2,627	12,729
2	Q3	3,330	9,339
3	Q4	2,316	4,507

BEFORE

AFTER

SQL as Task9_code9_1

```sql
-- D. Validate the final data
SELECT count(*) FROM BRONZE.LANDSLIDE_RAW;
```

	COUNT(*)
0	11,038

SQL as Task9_code9_2

```sql
SELECT count(*) FROM SILVER.FACT_LANDSLIDE;
```

	COUNT(*)
0	11,038

```sql
-- Check Gold (and/or check a specific metric you changed)
SELECT * FROM GOLD.QUARTERLY_LANDSLIDE_FATALITIES
ORDER BY QUARTER;
```

	QUARTER	NUM_LANDSLIDES	TOTAL_FATALITIES
0	Q1	2,760	4,486
1	Q2	2,630	12,737
2	Q3	3,332	9,351
3	Q4	2,316	4,507

Thank you!